

Nepal College of Information Technology (NCIT)Balkumari, Lalitpur | Affiliated to Pokhara University
Department of IT / Computer Engineering**LAB REPORT****Correlation Analysis and Hypothesis Testing**

Course: Data Science and Analytics (IT 405 / BUS 405)

Lab No.	04
Title	Correlation Analysis and Hypothesis Testing

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1. Objectives

On completion of this lab, the student will be able to:

- Measure the correlation between two numerical variables using Pearson, Spearman and Kendall coefficients.
- Measure the association between a numerical and a categorical variable using the point-biserial coefficient and the correlation ratio (eta).
- Measure the association between two categorical variables using the Chi-square test and Cramer's V.
- Apply basic hypothesis-testing methods — one-sample t-test, two-sample t-test, one-way ANOVA and the Chi-square test of independence — and interpret p-values.

2. Background Theory

Correlation measures the strength and direction of the relationship between two variables. A positive correlation means the variables move together; a negative correlation means one rises as the other falls; a value near zero means little linear/monotonic association. A key caution is that **correlation does not imply causation** — an observed association may be driven by a third (confounding) variable.

The appropriate measure depends on the **type** of the two variables being compared:

Numerical – Numerical	Pearson r	Linear association	-1 to +1
Numerical – Numerical	Spearman ρ	Monotonic (rank) association	-1 to +1
Numerical – Numerical	Kendall τ	Rank concordance	-1 to +1
Numerical – Categorical (binary)	Point-biserial r	Mean-difference association	-1 to +1

Numerical – Categorical (multi)	Correlation ratio η	Variance explained by groups	0 to 1
Categorical – Categorical	Cramer's V (via χ^2)	Strength of association	0 to 1

2.1 Hypothesis Testing Framework

A hypothesis test decides whether a pattern in the sample is strong enough to be considered real in the population. The standard steps are:

- **Null hypothesis (H_0):** there is no effect / no difference / no association.
- **Alternative hypothesis (H_1):** there is an effect / difference / association.
- **Significance level (α):** the risk of a false positive we accept, usually 0.05.
- **p-value:** the probability of observing data at least as extreme as ours if H_0 were true.
- **Decision rule:** if $p < \alpha$ we **reject H_0** (the result is statistically significant); otherwise we **fail to reject H_0** .

3. Tools and Libraries

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pandas	Loading data, .corr() matrix, crosstab / contingency tables, group statistics.
numpy	Numeric operations and the correlation-ratio computation.
scipy.stats	pearsonr, spearmanr, kendalltau, pointbisealr, ttest_1samp, ttest_ind, f_oneway, chi2_contingency.
matplotlib / seaborn	Scatter plots, heatmaps, box plots and grouped bar charts.

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy import stats

df = pd.read_csv("ncit_students.csv")
```

4. Dataset Description

The dataset records the academic profile of 80 NCIT students. It mixes numerical and categorical variables so that every type of correlation can be demonstrated.

StudentID	ID	Unique student identifier.

Gender	Categorical (binary)	Female / Male.
Faculty	Categorical (4 levels)	Computer, Software, Electronics, Civil.
StudyHours	Numerical	Average self-study hours per day.
Attendance	Numerical	Class attendance percentage.
InternetHours	Numerical	Recreational internet hours per day.
ExamScore	Numerical	Final exam score (0–100).
Result	Categorical (binary)	Pass (score ≥ 50) / Fail.

```
print(df.shape)
print(df.head())
```

Output (illustrative rows)

(80, 8)

```
StudentID Gender Faculty StudyHours Attendance InternetHours ExamScore Result S100 Female
Computer 6.1 86.6 3.6 66.4 Pass
S101 Male Software 4.7 73.0 5.5 52.0 Pass
S102 Female Computer 4.9 91.2 2.8 63.1 Pass
S103 Male Civil 3.2 70.4 4.1 41.7 Fail
S104 Female Electronics 5.5 88.7 4.9 55.8 Pass
```

5. Part A — Correlation between Numerical Variables

For two numerical variables we use three coefficients. **Pearson** captures linear association and assumes roughly normal data; **Spearman** works on ranks and captures any monotonic trend (robust to outliers and non-linearity); **Kendall** is also rank-based and measures the concordance of ordered pairs.

```
r, p1 = stats.pearsonr(df["StudyHours"], df["ExamScore"])
rho, p2 = stats.spearmanr(df["StudyHours"], df["ExamScore"])
tau, p3 = stats.kendalltau(df["StudyHours"], df["ExamScore"])
```

```
for col in ["Attendance", "InternetHours"]:
    rr, pp = stats.pearsonr(df[col], df["ExamScore"])
    print(col, round(rr,3), round(pp,4))
```

Output

```
Pearson (StudyHours, ExamScore): r = 0.512, p = 1.23e-06
Spearman (StudyHours, ExamScore): rho = 0.466, p = 1.35e-05
Kendall (StudyHours, ExamScore): tau = 0.325, p = 2.32e-05

Pearson (Attendance, ExamScore): r = 0.324, p = 0.0034
Pearson (InternetHours, ExamScore): r = -0.373, p = 0.0006
```

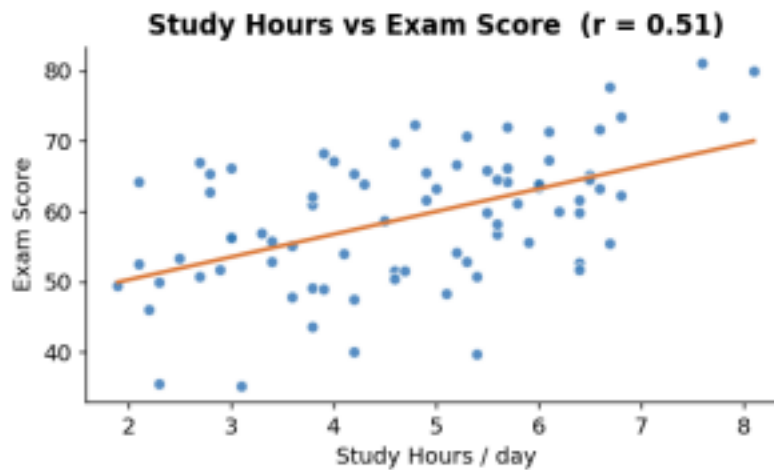


Fig. 1 — Study hours vs exam score with fitted line (Pearson $r = 0.51$).

Interpretation: Study hours show a **moderate-to-strong positive** correlation with exam score (all three coefficients agree and are highly significant). Attendance is a weaker positive driver, while internet hours show a **moderate negative** correlation — more recreational internet time is associated with lower scores.

5.1 Correlation Matrix and Heatmap

The pairwise Pearson correlations among all numerical columns are computed in one step and visualized as a heatmap.

```
num = ["StudyHours", "Attendance", "InternetHours", "ExamScore"]
print(df[num].corr().round(3))
sns.heatmap(df[num].corr(), annot=True, cmap="coolwarm", vmin=-1, vmax=1)
```

```
plt.show()
```

Output

```
StudyHours Attendance InternetHours ExamScore
StudyHours 1.000 0.116 -0.119 0.512
Attendance 0.116 1.000 -0.014 0.324
InternetHours -0.119 -0.014 1.000 -0.373
ExamScore 0.512 0.324 -0.373 1.000
```

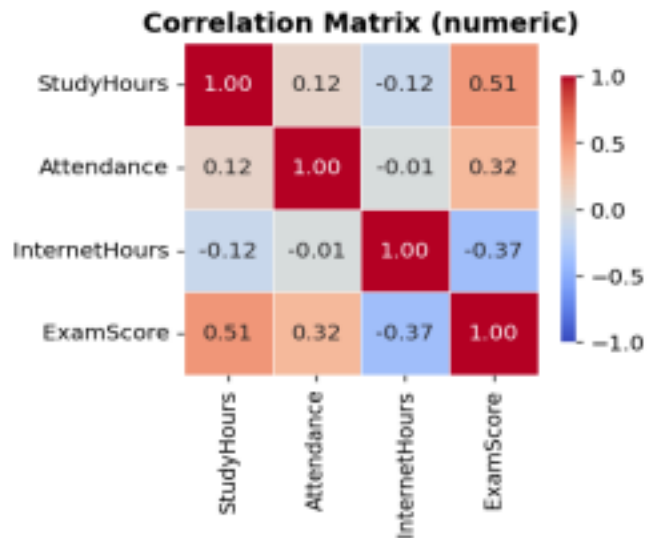


Fig. 2 — Correlation matrix heatmap of the numerical variables.

6. Part B — Correlation between Numerical and Categorical Variables

A categorical variable cannot be placed on a continuous axis, so Pearson is not appropriate. Two situations arise depending on the number of categories.

6.1 Binary Category — Point-Biserial Correlation

When the categorical variable has exactly **two** levels (here, Gender), it can be coded 0/1 and the **point-biserial** coefficient measures how strongly group membership relates to the numerical variable. It is mathematically equivalent to Pearson r on the coded variable.

```
df["Gender_bin"] = (df["Gender"] == "Female").astype(int)
r_pb, p = stats.pointbiserialr(df["Gender_bin"], df["ExamScore"])
print(round(r_pb,3), round(p,4))
print(df.groupby("Gender")["ExamScore"].agg(["count", "mean", "std"]).round(2))
```

Output

Point-biserial (Gender, ExamScore): $r_{pb} = -0.219$, $p = 0.0513$

	count	mean	std
Gender			
Female	36	56.93	10.40
Male	44	61.12	8.56

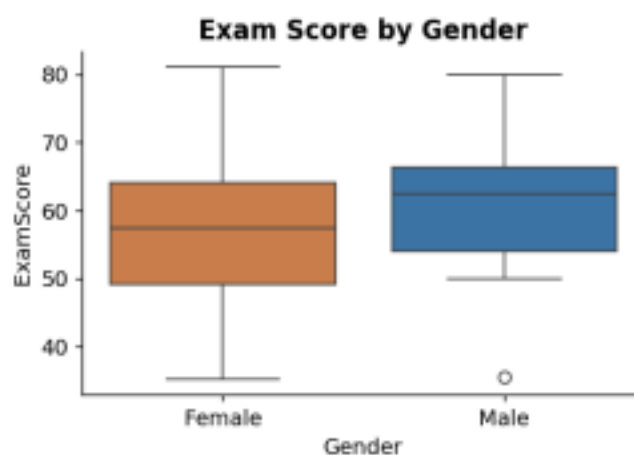


Fig. 3 — Exam score distribution by gender.

Interpretation: The point-biserial value (-0.22) is **weak** and its p -value (0.051) is just above 0.05 , so the gender–score association is **not statistically significant** at $\alpha = 0.05$. The two distributions overlap heavily — a good example of a borderline result.

6.2 Multi-Category — Correlation Ratio (Eta)

When the categorical variable has **more than two** levels (here, Faculty), the **correlation ratio** η measures the proportion of the numerical variable's variance that is explained by the groups. η^2 is the between-group sum of squares divided by the total sum of squares.

```
def correlation_ratio(categories, values):
    cats = pd.Categorical(categories); mean = values.mean()
```

```
ss_total = ((values - mean) ** 2).sum()
ss_between = sum(((values[cats == c].mean() - mean) ** 2) * (cats == c).sum() for c in
cats.categories)
return np.sqrt(ss_between / ss_total)
```

```
eta = correlation_ratio(df["Faculty"], df["ExamScore"])
print("eta =", round(eta,3), " eta^2 =", round(eta**2,3))
print(df.groupby("Faculty")["ExamScore"].agg(["count", "mean", "std"]).round(2))
```

Output

eta = 0.511 eta^2 = 0.261

	count	mean	std
Faculty			
Civil	9	49.36	6.01
Computer	24	64.29	7.09
Electronics	17	54.73	8.91
Software	30	60.71	9.58

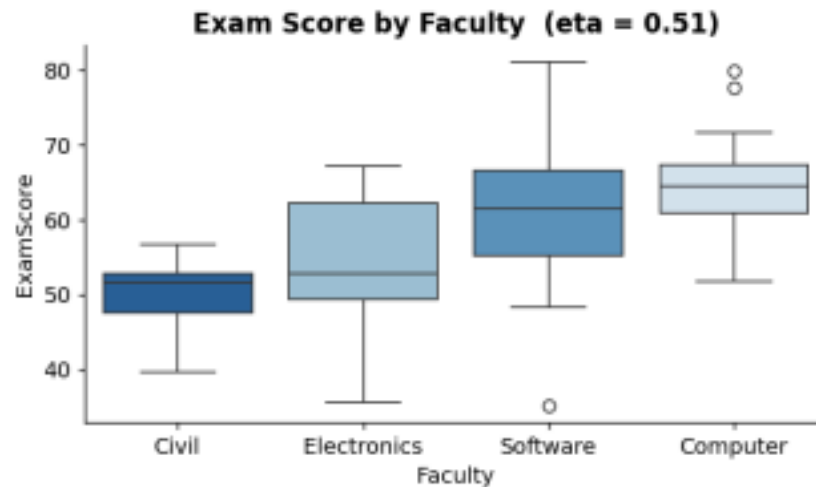


Fig. 4 — Exam score by faculty (eta = 0.51).

Interpretation: $\eta = 0.51$ indicates a **strong** association: about **26%** of the variation in exam scores is explained by faculty. Computer students score highest on average and Civil lowest — a difference confirmed by ANOVA in Part D.

7. Part C — Correlation between Categorical Variables

To assess the association between two categorical variables (Faculty and Result), we first build a **contingency table** (cross-tabulation) of counts, then apply the **Chi-square (χ^2) test of**

independence. The χ^2 statistic compares observed counts with the counts expected if the variables were independent. **Cramer's V** rescales χ^2 to a 0–1 strength measure.

```
ct = pd.crosstab(df["Faculty"], df["Result"])
print(ct)
chi2, p, dof, expected = stats.chi2_contingency(ct)

def cramers_v(conf):
    chi2 = stats.chi2_contingency(conf)[0]; n = conf.values.sum()
    r, k = conf.shape
    return np.sqrt(chi2 / (n * (min(r, k) - 1)))

print("chi2 =", round(chi2,3), " dof =", dof, " p =", round(p,4))
print("Cramer's V =", round(cramers_v(ct),3))
```

Output

```
Result Fail Pass
Faculty
Civil 4 5
Computer 0 24
Electronics 6 11
Software 3 27
```

```
chi2 = 15.305 dof = 3 p = 0.0016
Cramer's V = 0.437
```

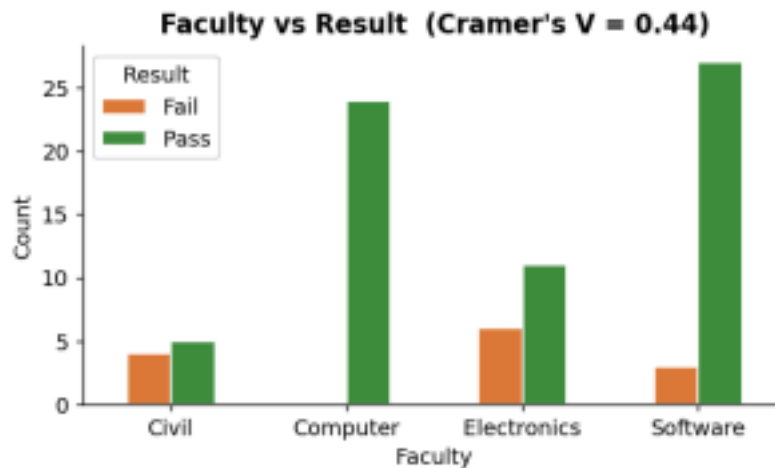
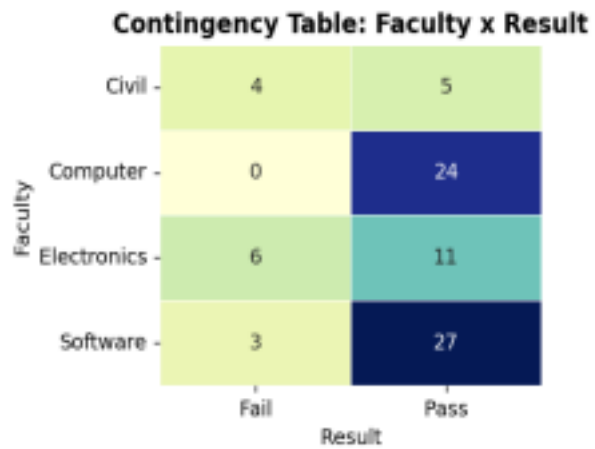


Fig. 5 — Pass/Fail counts by faculty (Cramer's V = 0.44).



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Fig. 6 — Contingency table heatmap (Faculty × Result).

Interpretation: Since $p = 0.0016 < 0.05$, we **reject** the null hypothesis of independence: Faculty and Result are **significantly associated**. Cramer's $V = 0.44$ indicates a moderate-to-strong relationship — Computer students pass at a much higher rate than Civil students.

8. Part D — Hypothesis Testing

Correlation tells us the strength of a relationship; hypothesis testing tells us whether an observed difference or association is statistically significant. Four common tests are demonstrated below, each at $\alpha = 0.05$.

8.1 One-Sample t-test

Tests whether the sample mean differs from a hypothesized value — here, whether the average exam score differs from a pass mark of 50.

```
t, p = stats.ttest_1samp(df["ExamScore"], 50)
print(round(df["ExamScore"].mean(),2), round(t,3), round(p,6))
```

Output

H0: mean ExamScore = 50 vs H1: mean != 50
mean = 59.23, t = 8.603, p = 5.79e-13 -> Reject H0

8.2 Two-Sample (Independent) t-test

Compares the mean exam scores of two independent groups (Female vs Male).

```
female = df[df["Gender"]=="Female"]["ExamScore"]
male = df[df["Gender"]=="Male"]["ExamScore"]
t, p = stats.ttest_ind(female, male, equal_var=False)
print(round(t,3), round(p,4))
```

Output

H0: mean(Female) = mean(Male) vs H1: means differ
Female = 56.93, Male = 61.12, t = -1.942, p = 0.0564 -> Fail to reject H0

8.3 One-Way ANOVA

Extends the t-test to compare the means of three or more groups — here, exam scores across the four faculties.

```
groups = [g["ExamScore"].values for _, g in df.groupby("Faculty")]
F, p = stats.f_oneway(*groups)
print(round(F,3), round(p,6))
```

Output

H0: all faculty mean scores are equal vs H1: at least one differs
F = 8.957, p = 3.75e-05 -> Reject H0

8.4 Chi-Square Test of Independence

Tests whether two categorical variables are associated (re-using the Faculty × Result table from Part C).

```
chi2, p, dof, expected = stats.chi2_contingency(ct)
print(round(chi2,3), dof, round(p,4))
```

Output

H0: Faculty and Result are independent vs H1: associated

$\chi^2 = 15.305$, $\text{dof} = 3$, $p = 0.0016 \rightarrow \text{Reject } H_0$

Write Summary of Tests and check which hypothesis are true?

One-sample t-test	Is mean ExamScore = 50?
Two-sample t-test	Female vs Male scores
One-way ANOVA	Scores across 4 faculties
Chi-square	Faculty vs Pass/Fail